Sample Spaces and Listing Outcomes

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS outcome *(putanga)* · sample space *(mokowā tīpako)* · systematic · equally likely *(tūpono ōrite)*

A Counting outcomes ●○○

Remember: The sample space is the set of all possible outcomes. When there are two stages, multiply the number of choices at each stage.

1. How many outcomes are there?

a) A coin is tossed _____ **b)** A dice is rolled _____

c) A coin is tossed twice _____ **d)** A coin is tossed and a dice is rolled _____

e) 2 shirts and 3 pairs of shorts _____ **f)** 4 sandwiches and 3 drinks _____

g) Two dice are rolled _____ **h)** A coin is tossed three times _____

B Listing outcomes ●●○

Remember: Be systematic: keep the first choice fixed and go through every second choice, then move on to the next first choice.

2. List all the possible outcomes. Use the letters and numbers given.

a) A coin is tossed twice (H, T).

b) A coin is tossed (H, T) and a spinner numbered 1, 2, 3 is spun.

c) You choose a pizza (cheese C, ham H, vegetable V) and a drink (juice J, water W).

d) A spinner with the letters A, B, C is spun twice.

C Sample space tables ●●○

Remember: A table shows every outcome of two events once. Each cell is one outcome.

3. A spinner numbered 1 to 4 and a spinner numbered 1 to 3 are spun, and the two numbers are added. Complete the table to show every possible total.

+	1	2	3	4
1				
2				
3				

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4. Use your table.

a) How many outcomes are there altogether? _____

b) What is the largest possible total? _____

c) How many outcomes give a total of 5? _____

d) How many outcomes give a total of less than 4? _____

D Probabilities from a sample space ●●●

> **Remember:** Count the outcomes that fit, then divide by the total number of outcomes.

5. Use your table for the two spinners. Find the probability that the total is:

a) 5 **b)** 2 **c)** an even number **d)** more than 5

6. A coin is tossed twice. Find the probability of getting:

a) two tails **b)** two heads **c)** one head and one tail, in either order

d) at least one tail

7. A coin is tossed and a dice is rolled. Find the probability of getting:

a) a head and a 6 **b)** a tail and an even number **c)** a head and a number less than 3

8. Two spinners numbered 1 to 3 are spun. Find the probability that both show the same number.

★ Extension

9. A coin is tossed three times. List the 8 outcomes first, then find the probability of getting:

a) three heads **b)** exactly two heads **c)** at least one head

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Sample Spaces and Listing Outcomes

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS outcome *(putanga)* · sample space *(mokowā tīpako)* · systematic · equally likely *(tūpono ōrite)*

A Counting outcomes ●○○

Remember: The sample space is the set of all possible outcomes. When there are two stages, multiply the number of choices at each stage.

1. How many outcomes are there?

a) A coin is tossed _____

b) A dice is rolled _____

c) A coin is tossed twice _____

d) A coin is tossed and a dice is rolled _____

e) 3 shirts and 2 pairs of shorts _____

f) 5 sandwiches and 3 drinks _____

g) Two dice are rolled _____

h) A coin is tossed three times _____

B Listing outcomes ●●○

Remember: Be systematic: keep the first choice fixed and go through every second choice, then move on to the next first choice.

2. List all the possible outcomes. Use the letters and numbers given.

a) A coin is tossed twice (H, T).

b) A coin is tossed (H, T) and a spinner numbered 1, 2, 3, 4 is spun.

c) You choose an ice cream (vanilla V, chocolate C, mint M) and a topping (nuts N, sauce S).

d) A spinner with the letters X, Y, Z is spun twice.

C Sample space tables ●●○

Remember: A table shows every outcome of two events once. Each cell is one outcome.

3. A spinner numbered 1 to 4 and a spinner numbered 1 to 4 are spun, and the two numbers are added. Complete the table to show every possible total.

+	1	2	3	4
1				
2				
3				
4				

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4. Use your table.

a) How many outcomes are there altogether? _____

b) What is the largest possible total? _____

c) How many outcomes give a total of 5? _____

d) How many outcomes give a total of less than 4? _____

D Probabilities from a sample space ●●●

> **Remember:** Count the outcomes that fit, then divide by the total number of outcomes.

5. Use your table for the two spinners. Find the probability that the total is:

a) 5 **b)** 2 **c)** an even number **d)** more than 5

6. A coin is tossed twice. Find the probability of getting:

a) two tails **b)** two heads **c)** one head and one tail, in either order

d) at least one tail

7. A coin is tossed and a dice is rolled. Find the probability of getting:

a) a head and a 6 **b)** a tail and an even number **c)** a head and a number less than 5

8. Two spinners numbered 1 to 3 are spun. Find the probability that both show the same number.

★ Extension

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